

Cloud Native Data Pipeline Platforms for AI

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Aug 2026

Meeting Objectives

“As AI moves from frontier labs into mainstream production, the [operational challenges](#) start to look increasingly cloud native: orchestration, autoscaling, routing, security, policy, and observability.”

[Jonathan Bryce](#), Exec Director, CNCF at KCD Texas, May 2026

1. Build on infrastructures – CNCF, LinuxFoundation, and open-source & community reference for sharing
2. Feedback shaping direction – challenges and use cases (*via offline or Github site*)
3. CNCF engagement – TAG, paper, developer community?

Introductions



Pradeep Kalluri

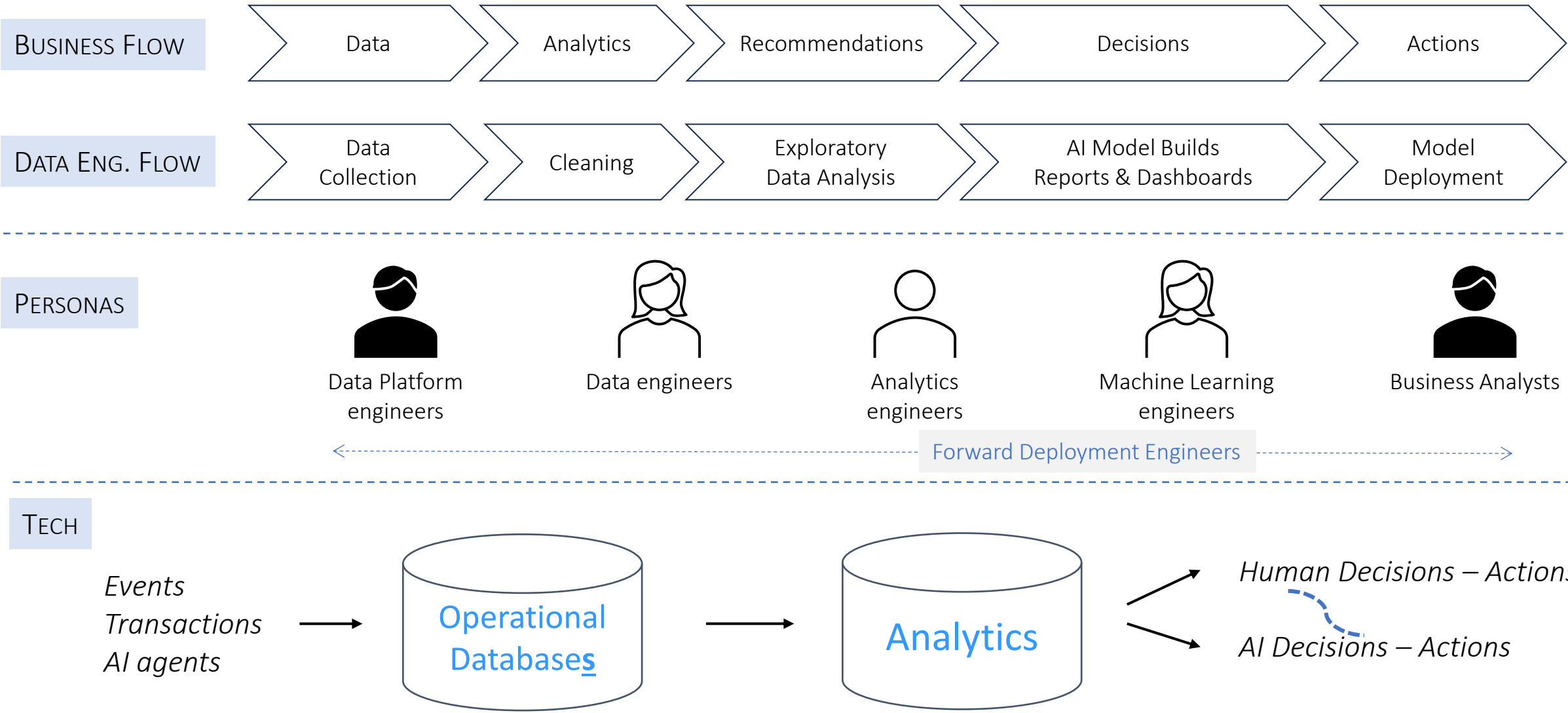
- Data Engineer with experience building production data platforms at NatWest, Accenture, and Dpoint
- Specialized in cloud-native architectures, real-time processing with Kafka and Spark, and data quality frameworks
- Published technical writer on Medium, sharing practical lessons from production systems
- Passionate about making data platforms reliable and trustworthy
- <https://www.linkedin.com/in/pradeepkalluri/>



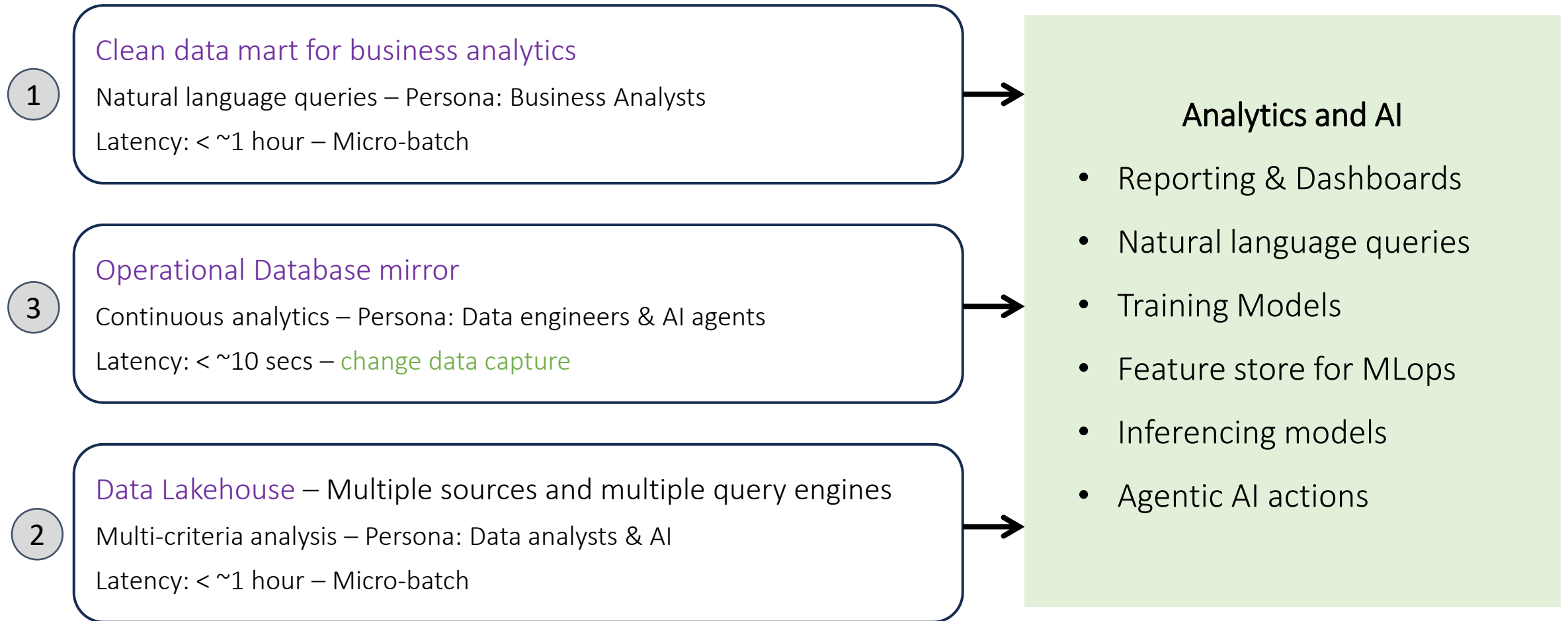
Raghu Shankar

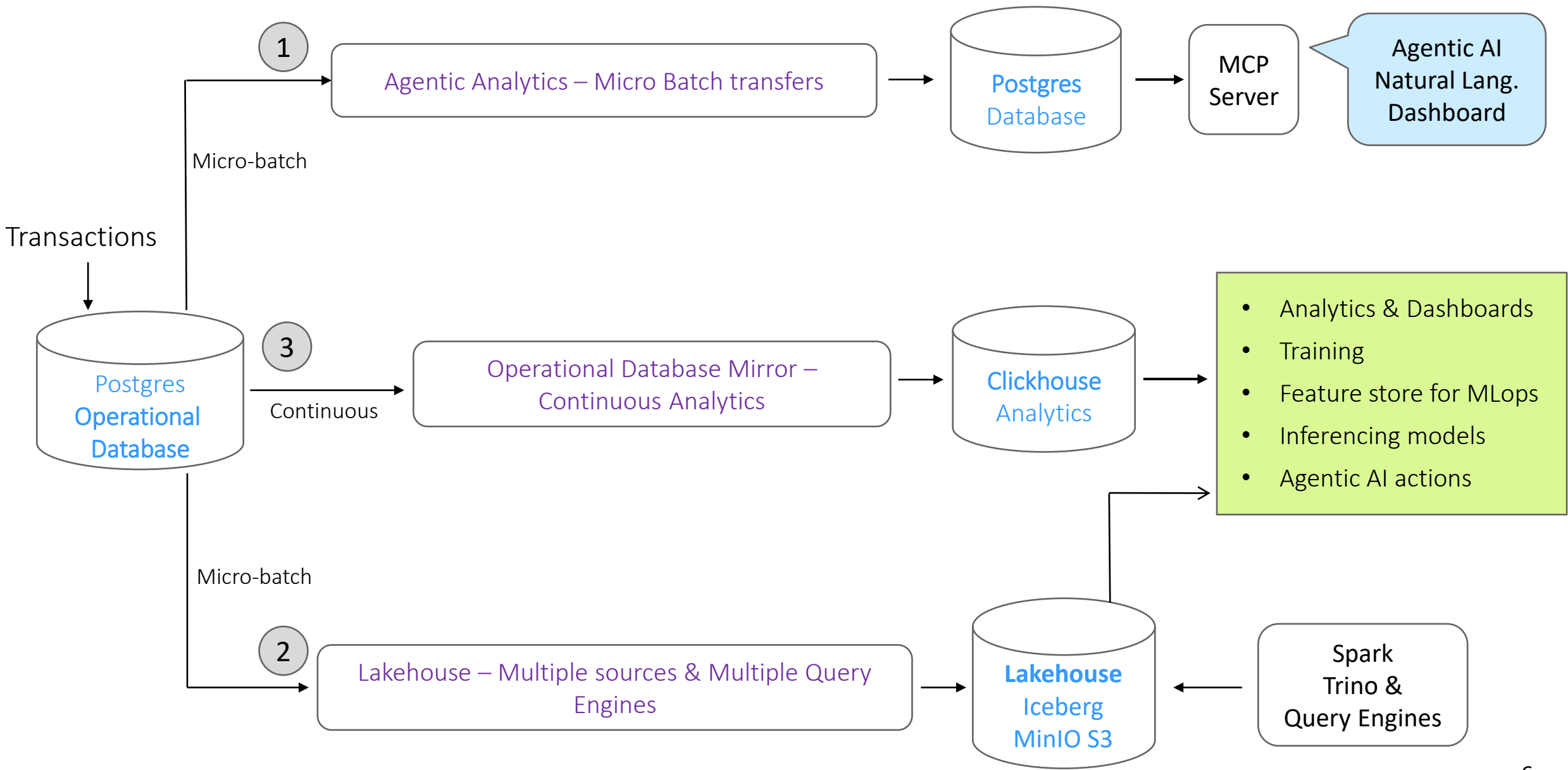
- Entrepreneur: cloud and Semiconductor systems
- Dell: Server product management – HPC & AI
- Intel: Architect and Product Management
- <https://www.linkedin.com/in/raghu-shankar/>

Data to Actions cycle shrinking rapidly in human-AI era

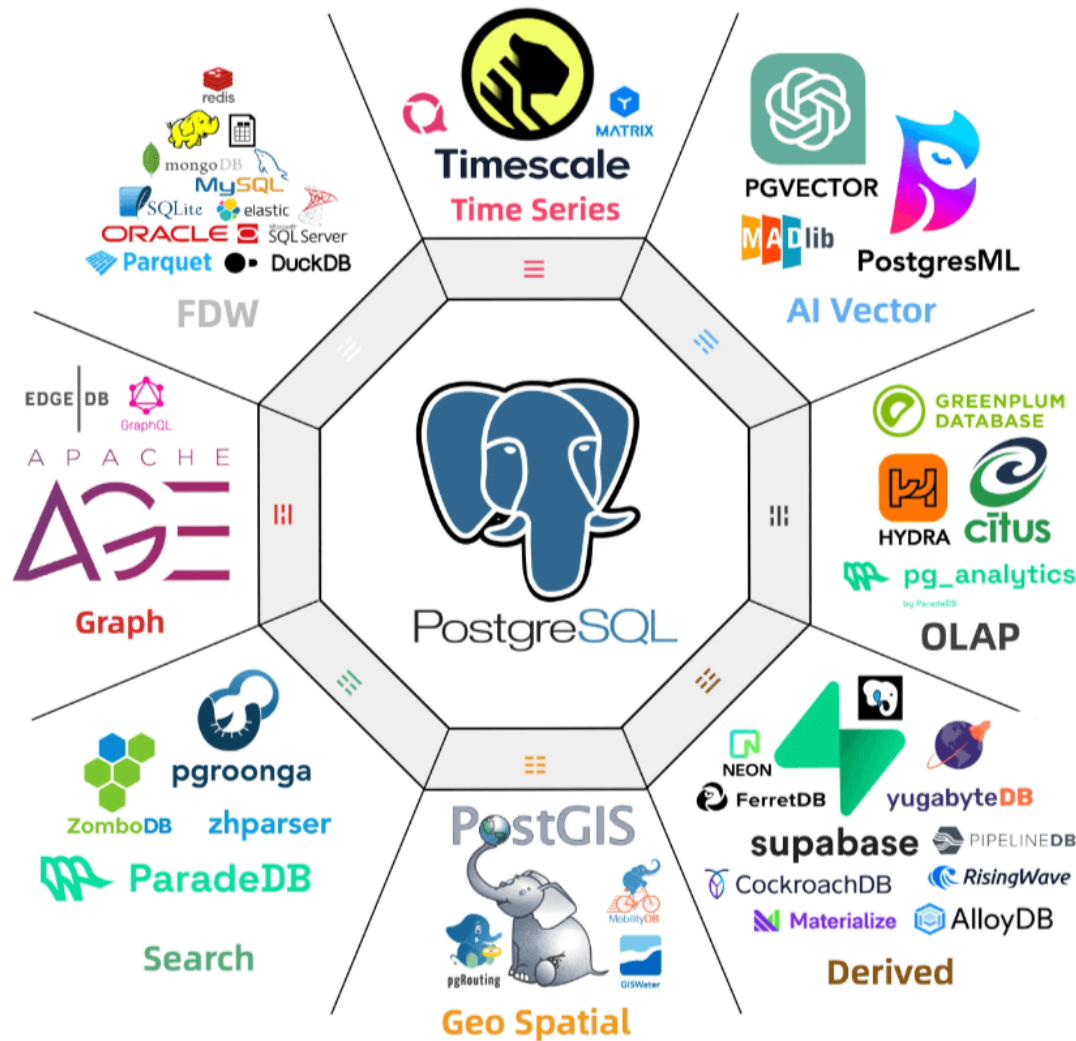


Data quality and data urgency drives high confidence & timely decisions





Postgres extension framework: Foundational database platform



Top 5 Database

Open-source

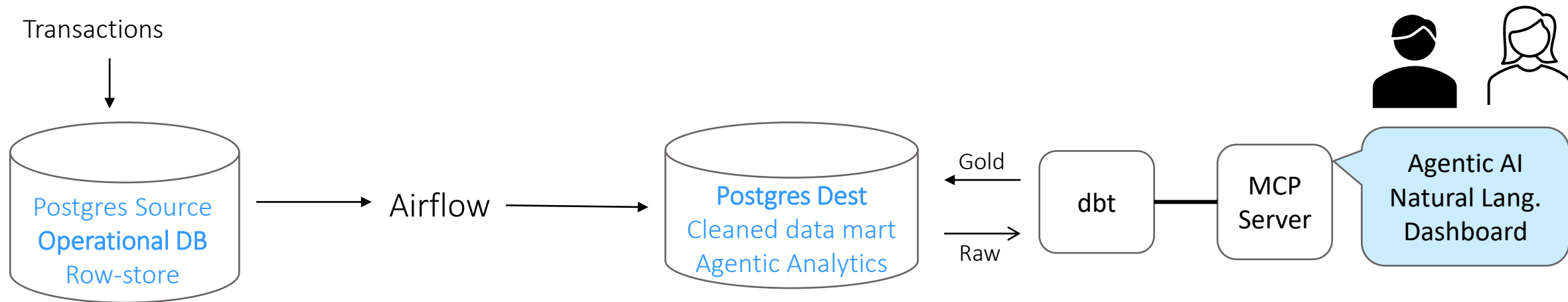
Widely deployed

Investment protection

Source:

[DB-Engines Ranking - popularity ranking of database management systems](#)

Pipe #1: Agentic Analytics – Incremental Micro-Batch Data Transfers



Postgres Source

- Postgres pod
- Exporter pod metrics
- Persistent volume – managed by K8s
- [Grafana dashboard - 9628](#)

Airflow

Create, schedule and monitor workflows with directed acyclic graphs (DAG). Scheduling, retries, and alerting

- Helm
- NodePort UI – secure login

- API server • DAG processor
- Scheduler • Statsd
- Postgres – stores metadata persistent volume (PV)
- Triggerer logs in PV

Postgres Destination

- Postgres pod
- Exporter pod metrics
- Persistent volumes - managed by K8s
- [Grafana dashboard - 9628](#)

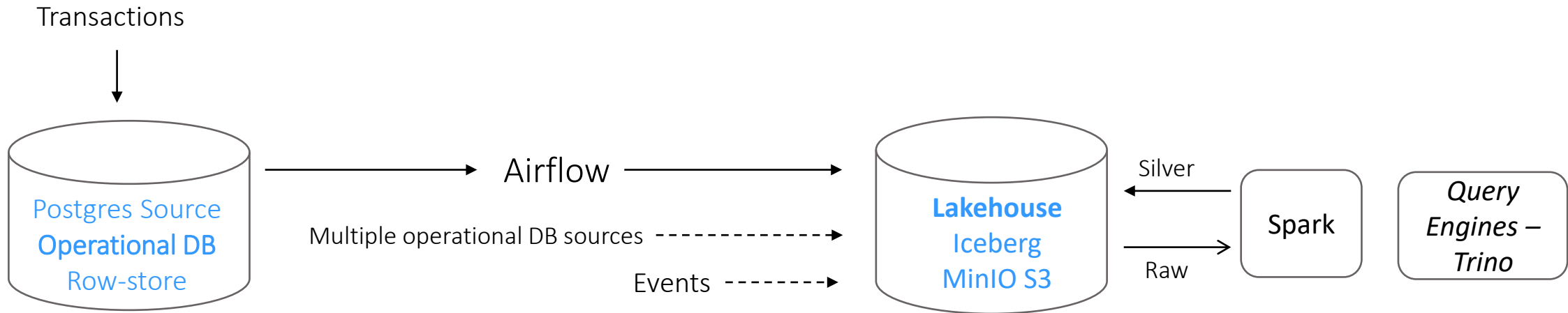
Data Build Tool

- Lightweight framework to write, test, and run [SQL transformations](#) inside Postgres
- Bronze → Silver → Gold
- Executed on demand by Airflow

Natural Lang queries - MCP server

- Reads SQL – zero hallucination
- Autonomous validation dbt test
- Human in the loop for code commits
- NodePort for UI – secure login

Pipe #2: Lakehouse – Multi sources & Multiple Query Engines



MinIO S3

- Object storage – huge capacity/\$
- MinIO **pod** includes Iceberg tables
- Persistent Volume via Rancher
- NodePort – Data buckets
- [Grafana Dashboard 13502](#)

Apache Iceberg

- Iceberg: ACID, versioning, auditing, data quality and lineage
- Supported by major vendors
- Data formats: Parquet, ORC, or Avro
- Ingest data from many sources – streaming or batch
- Many query engines to read, write without breaking data consistency [via Iceberg connectors](#)

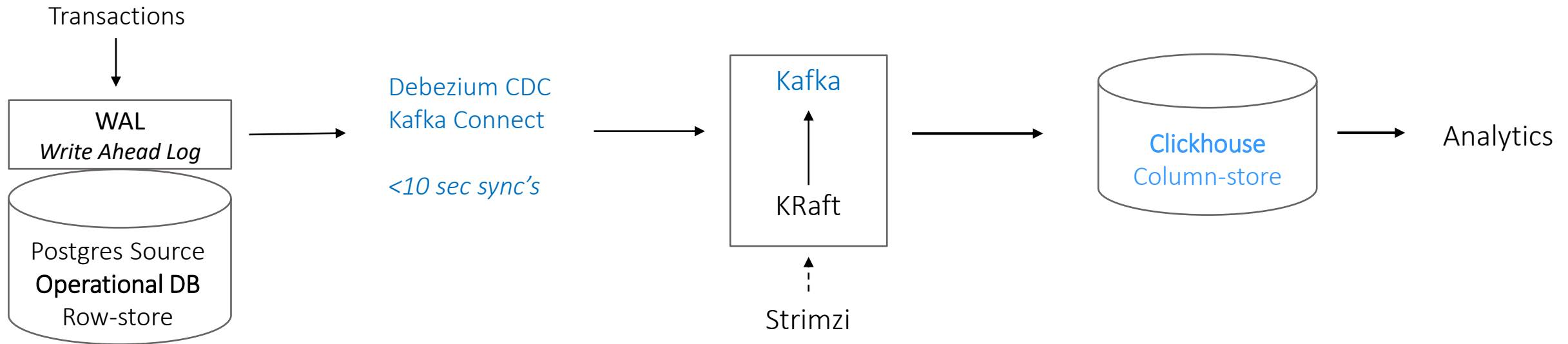
Apache Spark

- Read/write data in compressed Parquet
- Bronze → Silver
- Scale up for massive (100GB) data sets
- 1x Master pod and 2x worker pods – processing in memory (no PVC)
- NodePort for UI (secure login)

Trino distributed Query engine

- Helm chart
- Native Iceberg connector – shares JDBC catalog with Spark – data consistency
- 1x co-ordinator and 2x worker pods

Pipe #3: Operational Database Mirror – Continuous Analysis



Debezium – Kafka Connect

- Connector to source database
- Read write-ahead log (WAL) for changes
- Changes → events streams to Kafka
- State stored in Kafka topics
- Connect pod

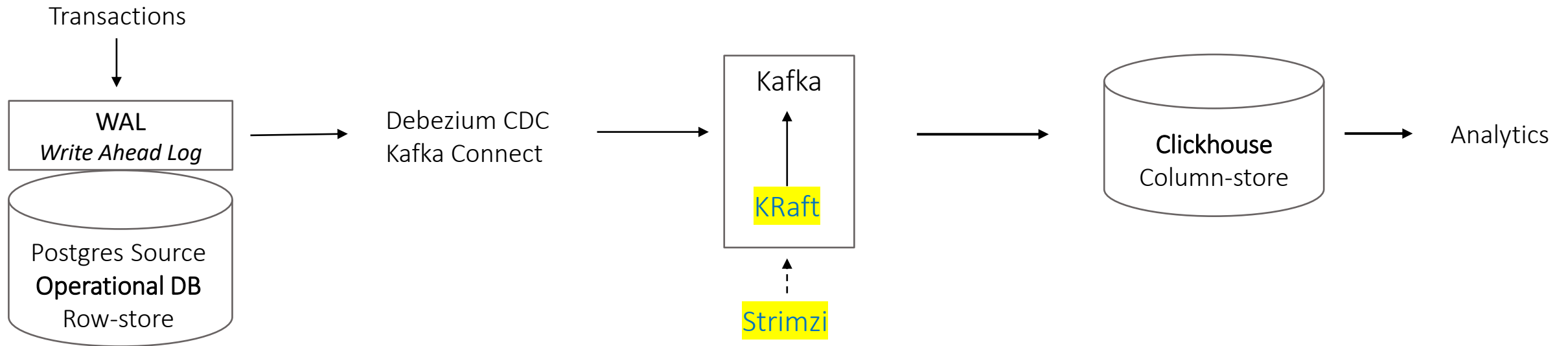
Apache Kafka

- Event streaming platform – massive message queue
- Topics and partitions (scalability)
- Resiliency, idempotency and fault-tolerant
- Pod includes Debezium
- NodePort for UI – topics, brokers, and consumers - POD

ClickHouse

- Column-oriented database built for real-time analytics
- Kafka connects to built-in Kafka engine
- Hybrid transaction and analytics processing (HTAP) - “basic”
- Pod and PVC

Pipe #3: Operational Database Mirror – Continuous Analysis



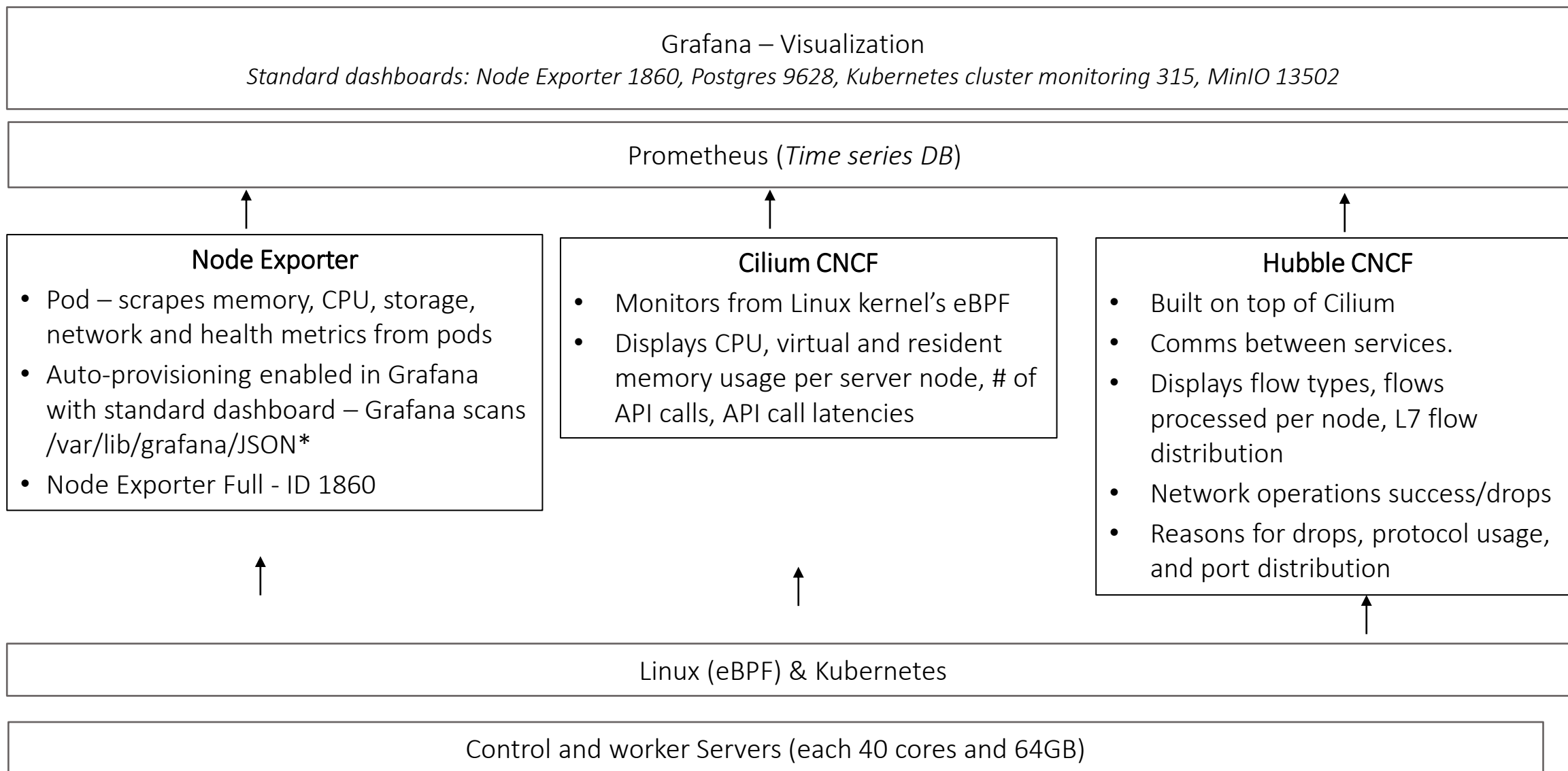
Strimzi CNCF

- Runs Kafka cluster on Kubernetes in various deployment configurations
- Kubernetes operator pattern to deploy, run, and manage collections of Kafka brokers
- Declarative way to manage Kafka clusters – Strimzi handles deployment, upgrades, and maintenance

KRaft

- Built-in metadata management system,
- Replaces Zookeeper
- KRaft provides more flexibility to configure Kafka server roles
- Kafka configured as combined broker and controller
- Raft based protocol manages metadata more reliably and efficiently

Platform and Observability



Security: Built in from Day One – continuously strengthened for production

- Secrets
 - Generated secrets per deployment (via deploy script)
 - Everything reads from the etl-secrets Secret, nothing hardcoded;
 - Spark passwords passed via task env so they never reach Airflow's metadata DB or task logs
- AI dashboard, three independent layers
 - LLM output gated to a single SELECT;
 - Dedicated SELECT-only role;
 - [Read-only](#) enforced server-side in all three stores
- Supply chain —
 - gitleaks, bandit, pip-audit,
 - npm audit,
 - kubeconform: Continuous Integration (CI) 16 checks on every Pull Request (PR)

<https://medium.com/@raghu.v.shankar/data-pipelines-platform-for-analytics-and-ai-proof-of-concept-poc-bring-your-own-use-case-6bfbb8e9aa2e>



Raghu V Shankar
Product management and engineering professional. Entrepreneur incubating cloud native and chiplet architectures - <https://www.linkedin.com/in/raghu-shankar/>

AI + Cloud Native + PostgreSQL + Data Engineering ✓ Proof Of Concept POC ✓

Data Pipelines Platform for Analytics and AI: Proof of concept (POC) — Bring your own use case



Raghu V Shankar · 23 min read · Jul 21, 2026

3

Data Quality and Decision Urgency of the use case drive pipeline architectures for high-quality and timely decision making

Pradeep Kalluri and Raghu Shankar

July 2026

Summary

Data is the foundation for many of decisions made collaboratively by humans and AI. Data quality and decision urgency are becoming more critical as more decisions are transitioned to AI. This paper and associated proof of concept (POC) platform covers three independent data pipeline architectures to deliver high quality data at required levels of urgency for enterprises tackling data pipelines for human and AI collaborative analytics, decisions and actions. This POC is built on proven open-source and cloud-native infrastructures including Kubernetes, Docker, Cilium, PostgreSQL, ClickHouse, Apache Iceberg, MinIO object store S3, Apache Airflow, data build tool (dbt), natural language queries via MCP, Spark, Debezium, Apache Kafka, Strimzi, KRaft, Prometheus, and Grafana.

<https://github.com/kalluripradeep/modern-etl-stack>

Issues: 40+ PR's: 140+

Issues 2 Pull requests Actions Projects Security and quality Insights

modern-etl-stack Public

main 82 Branches 0 Tags

Go to file Add file Code

Merge pull request #143 from kalluripradeep/docs/storage-growth 38efbe · 4 days ago 295 Commits

.github	fix(monitors): query the metric names statsd-exporter actu...	2 weeks ago
airflow/dags	fix: reclaim disk automatically instead of asking people to re...	5 days ago
dbt	feat: add the gold layer — dimensional star schema	last month
docker	fix(build): allow the Spark JARs to be supplied locally	last week
docs	fix: bug sweep — spark bronze paths, dashboard grants, deci...	last month
k8s	fix(airflow): stop #140 reserving memory and cpu it did not n...	4 days ago
monitoring	fix(monitors): query the metric names statsd-exporter actu...	2 weeks ago
sample-data	fix: make Kubernetes observability actually work + cleanups	last month
scripts	fix(monitors): query the metric names statsd-exporter actu...	2 weeks ago
spark/jobs	fix: reclaim disk automatically instead of asking people to re...	5 days ago
ui	chore(ui): bump nanoid past GHSA-2v37-7h3g-55p8	last week
.env.example	feat: ClickHouse columnar mirror fed from the CDC topics	last month
.gitattributes	fix: resolve lint-shell and npm audit CI failures	last month
.gitignore	fix(build): allow the Spark JARs to be supplied locally	last week
.markdownlint-cli2.jsonc	docs: clean up markdown lint so the advisory check stops fla...	last month
.yamllint	ci: fix mypy module shadowing and expand pipeline to 15 ch...	last month
DEPLOY_GUIDE.md	fix: reclaim disk automatically instead of asking people to re...	5 days ago

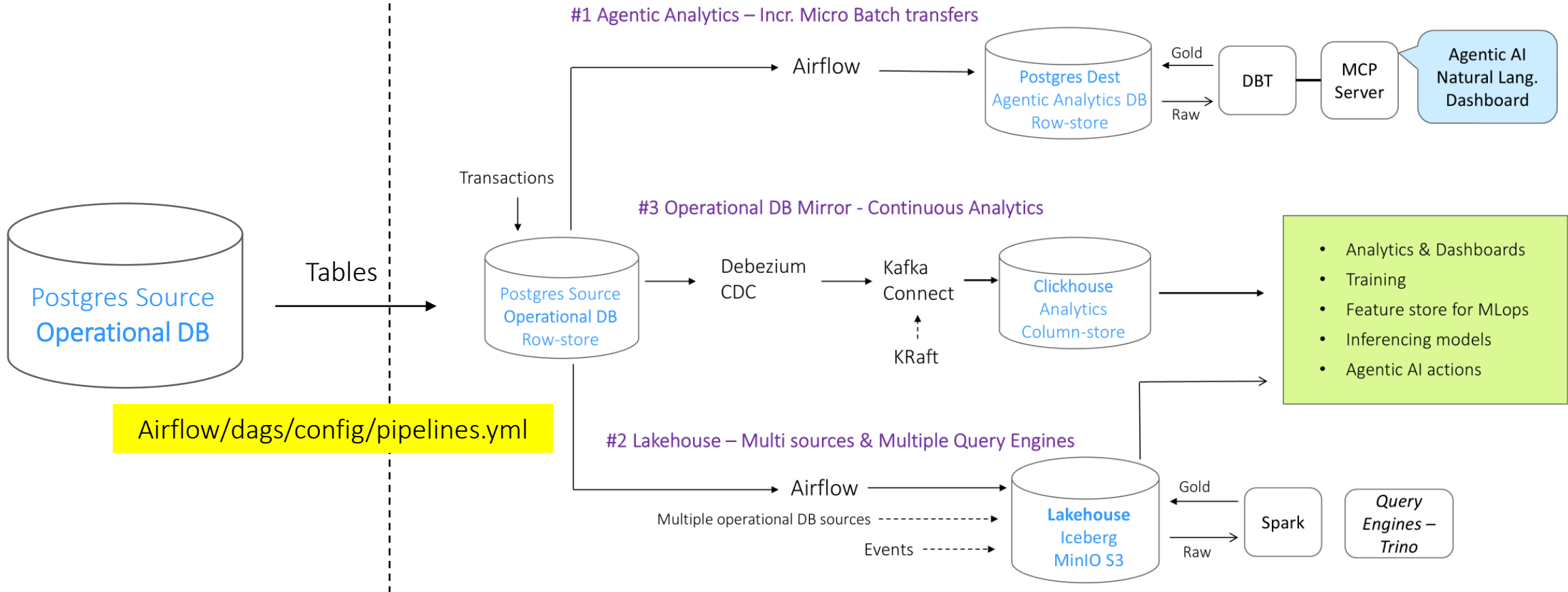
Deploy Guide & Script

PROD ENV

SANDBOX

1. Deploy script – deploys all 3 pipes
2. Runs one e-commerce data transfer test
3. Liveness Traffic Test – continuous data stream
4. Ingestion pipeline DAG

5. Extensive data tests
6. Benchmarks, tpc-c



Observability – Dashboards

Airflow top level

Home

Dags

Assets

Browse

Admin

Security

Dags

Dag Runs

Task Instances

Search Dags

Ctrl+K

o*

All

Failed

Queued

Running

Success

Required Actions

All

Active

Paused

Filter by tag

All

Favorite

Unfavorite

Sort by Display Name (A-Z)

2 Dags

ingest_source_to_bronze

bronze, etl, multi-table

toggle

play

star

trash

Schedule

Latest Run

Next Run

0 *****

2026-08-14 19:00:00

spark_transform_silver

compaction, iceberg, scalability, +2 more

toggle

play

star

trash

Schedule

Latest Run

Next Run

0 *****

2026-08-14 19:00:00

Airflow – Ingest source to bronze - Runs list

Dag
ingest_source_to_bronze

Search Dags Ctrl+K Trigger

Filter

3d20:15:34

1d22:07:47

ingest_customers
ingest_order_items
ingest_orders
ingest_products
trigger_spark_pipeline
dbt_transformations ^
customers_clean_run
dim_customer v
order_items_clean_run
orders_clean v
products_clean v
v_customers_run
v_daily_revenue v
v_orders_run
v_products_run
dim_date v
dim_product v
fact_order_items v
emit_data_quality_metrics
prune_bronze

ingest_source_to_bronze

Schedule Latest Run Next Run Max Active Runs Owner Tags Latest Dag Version

0 **** 2026-08-14 19:00:00 1 of 1 data_engineering bronze, etl, multi-table v2

Overview Runs Tasks Calendar Backfills Audit Log Code Details

Filter

☐	Dag Run ID	Run After	State	Run Type	Triggering User	Start Date	Duration	Dag Version(s)
☐	scheduled_2026-08-15T00:00+00:00	2026-08-14 19:00:00	Running	Scheduled		2026-08-14 19:00:00		v2
☐	scheduled_2026-08-14T23:00+00:00	2026-08-14 18:00:00	Failed	Scheduled		2026-08-14 18:00:00	00:12:43	v2
☐	scheduled_2026-08-14T22:00+00:00	2026-08-14 17:00:00	Failed	Scheduled		2026-08-14 17:00:00	00:47:44	v2
☐	scheduled_2026-08-14T21:00+00:00	2026-08-14 16:00:00	Success	Scheduled		2026-08-14 16:00:00	00:31:19	v1
☐	scheduled_2026-08-14T20:00+00:00	2026-08-14 15:00:00	Success	Scheduled		2026-08-14 15:15:57	00:43:09	v1
☐	scheduled_2026-08-14T19:00+00:00	2026-08-14 14:00:00	Success	Scheduled		2026-08-14 14:00:00	01:15:56	v1
☐	scheduled_2026-08-14T18:00+00:00	2026-08-14 13:00:00	Success	Scheduled		2026-08-14 13:00:01	00:40:17	v1
☐	scheduled_2026-08-14T17:00+00:00	2026-08-14 12:00:00	Success	Scheduled		2026-08-14 12:00:00	00:22:24	v1
☐	scheduled_2026-08-14T16:00+00:00	2026-08-14 11:00:00	Success	Scheduled		2026-08-14 11:00:00	00:29:45	v1
☐	scheduled_2026-08-14T15:00+00:00	2026-08-14 10:00:00	Success	Scheduled		2026-08-14 10:00:01	00:26:33	v1

Spark



Spark Master at spark://spark-master-0.spark-master.etl.svc.cluster.local:7077

URL: spark://spark-master-0.spark-master.etl.svc.cluster.local:7077

Alive Workers: 2

Cores in use: 4 Total, 0 Used

Memory in use: 4.0 GiB Total, 0.0 B Used

Resources in use:

Applications: 0 [Running](#), 182 [Completed](#)

Drivers: 0 Running, 0 Completed

Status: ALIVE

▼ Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260814204603-10.0.1.52-46409	10.0.1.52:46409	ALIVE	2 (0 Used)	2.0 GiB (0.0 B Used)	
worker-20260814204606-10.0.1.63-37763	10.0.1.63:37763	ALIVE	2 (0 Used)	2.0 GiB (0.0 B Used)	

▼ Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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▼ Completed Applications (182)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260815000126-0281	Orders-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/15 00:01:26	airflow	FINISHED	3 s
app-20260814233106-0280	Customers-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/14 23:31:06	airflow	FINISHED	13 s
app-20260814232416-0279	Customers-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/14 23:24:16	airflow	FINISHED	6 s
app-20260814231830-0278	Orders-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/14 23:18:30	airflow	FINISHED	14 s
app-20260814225333-0277	Orders-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/14 22:53:33	airflow	FINISHED	14 s
app-20260814223429-0276	OrderItems-Bronze-to-Silver-Incremental	4	2.0 GiB		2026/08/14 22:34:29	airflow	FINISHED	17 s

MinIO – Bronze and Silver buckets

MINIO

OBJECT STORE

Community Edition

+

Create Bucket

Q

Filter Buckets

Buckets

airflow-logs

bronze

silver

Object Browser

Q

Start typing to filter objects in the bucket

?

☾

bronze

Created on: Tue, Jul 28 2026 09:22:53 (CDT) Access: PRIVATE 2.4 MiB - 643 Objects

Rewind

Refresh

Upload

<

bronze

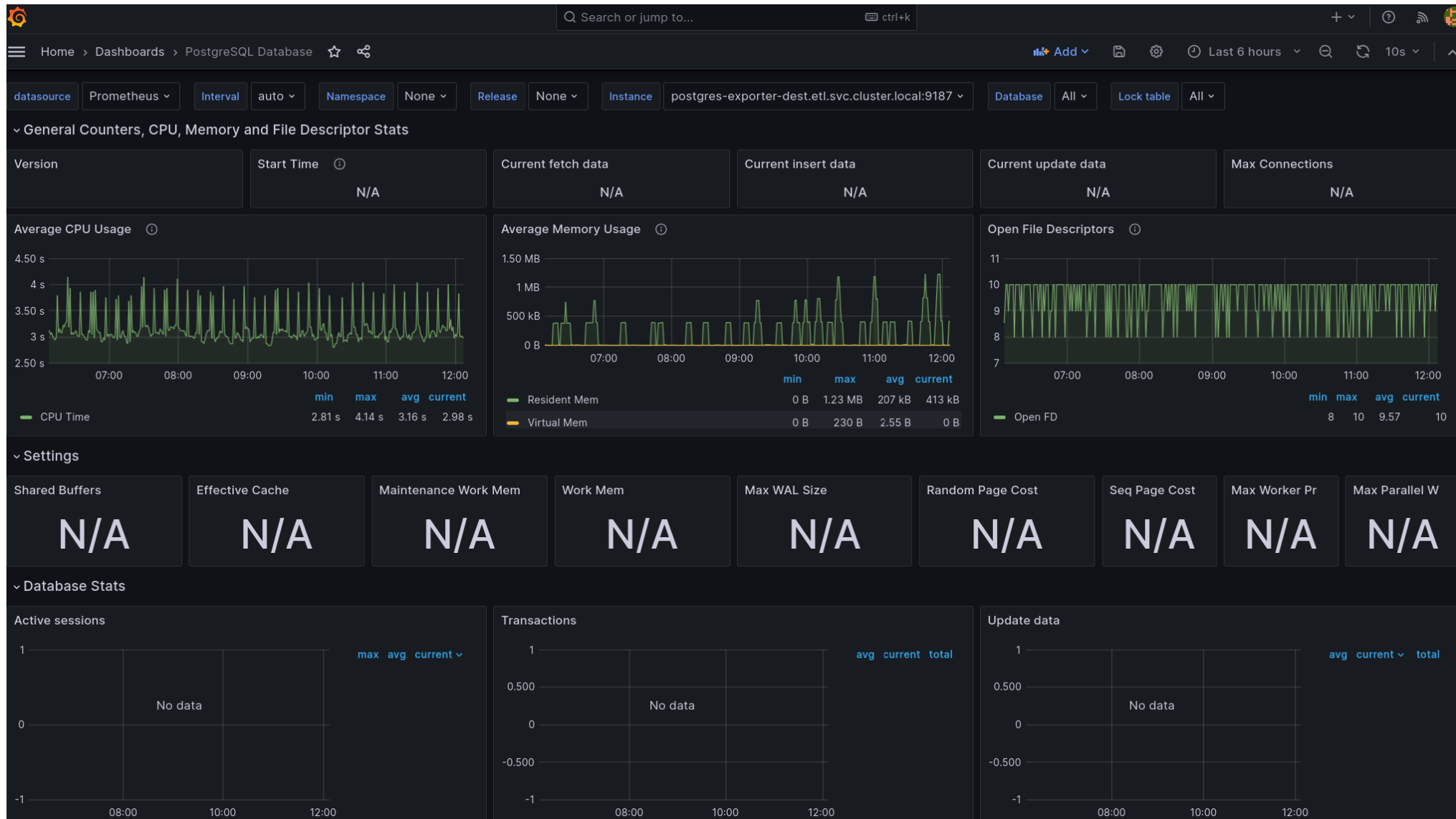
Create new path

Apache Kafka dashboard

The screenshot shows the Apache Kafka dashboard interface. At the top left, it says "UI for Apache Kafka" and "83b5a60 v0.7.2". The left sidebar has a "Dashboard" tab selected, and below it, a list of items: "local", "Brokers", "Topics", and "Consumers". The main content area is titled "Dashboard" and features two summary cards: "Online" with "1 clusters" and "Offline" with "0 clusters". Below these cards is a toggle switch labeled "Only offline clusters" which is currently turned off. A table follows, displaying details for the "local" cluster. The table has columns for Cluster name, Version, Brokers count, Partitions, Topics, Production, and Consumption. The data row shows "local" as the cluster name, "3.5-IV2" as the version, "1" as the brokers count, "0" for both partitions and topics, and "0 Bytes" for both production and consumption.

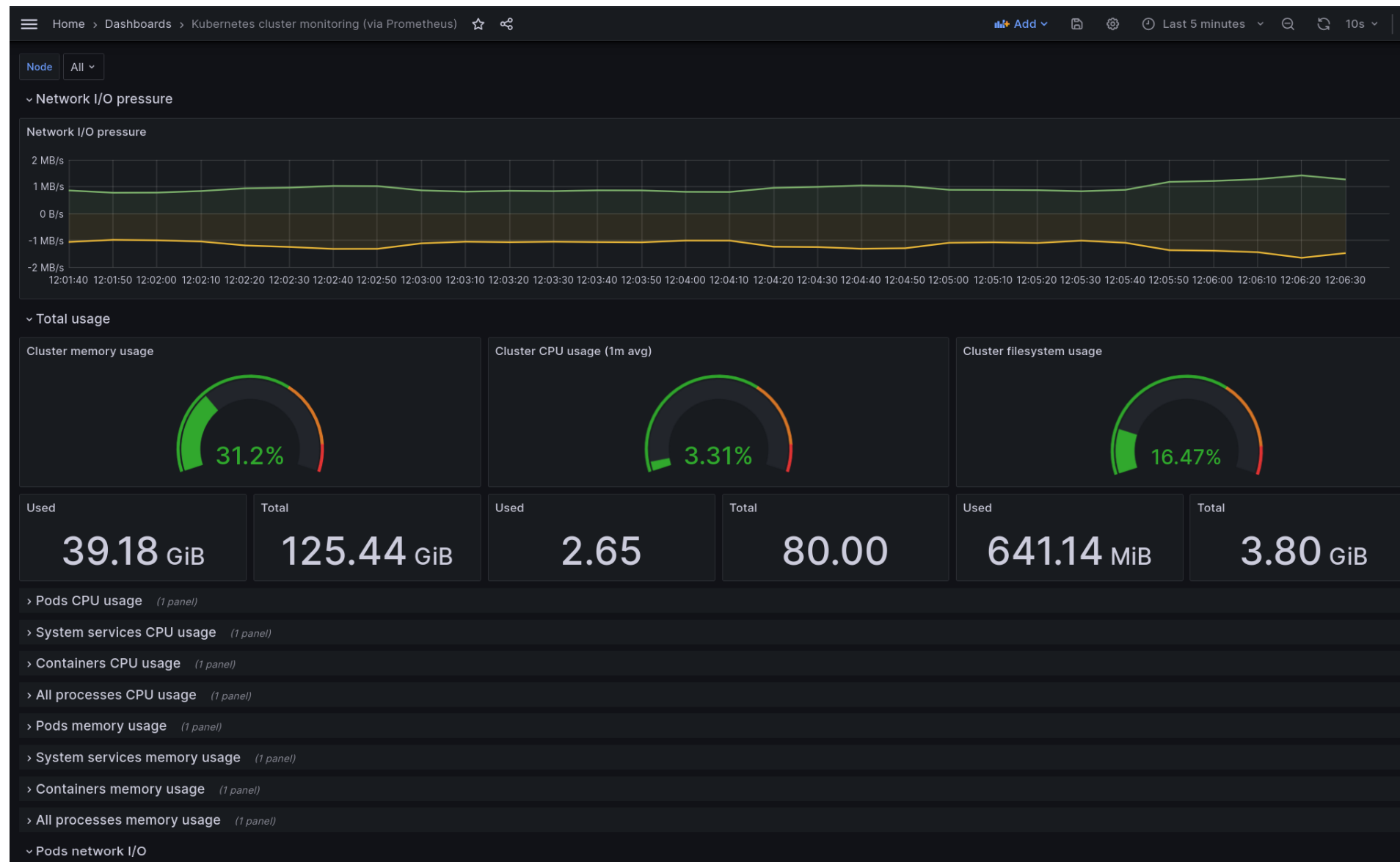
Cluster name	Version	Brokers count	Partitions	Topics	Production	Consumption
local	3.5-IV2	1	0	0	0 Bytes	0 Bytes

Grafana dashboard - 9628: PostgreSQL — open file descriptors, CPU/mem usage



Kubernetes cluster monitoring – Network and CPU/memory usage, pods, ..

<https://grafana.com/grafana/dashboards/1860-node-exporter-full/>



Roadmap

Roadmap vision: Extend Data to Actions architecture

BUSINESS FLOW

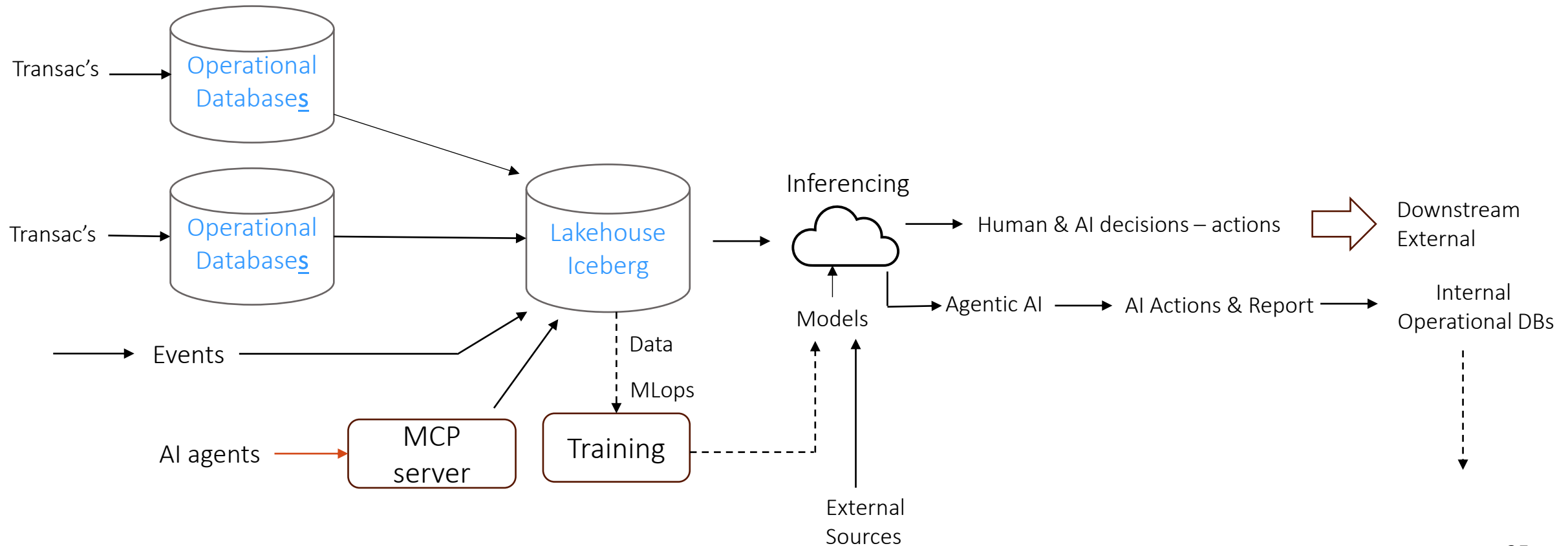
Data

Analytics

Recommendations

Decisions

Actions



References & Related Readings – page #1

1. “From Raw to Refined: Data Pipeline Architecture at Scale”, Pradeep Kalluri via Medium, Nov 2025
2. “Batch vs Streaming – Choosing the right architecture”, Pradeep Kalluri, LinkedIn, Feb 2026
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4. “Introducing the dbt MCP Server – Bringing Structured Data to AI Workflows and Agents”, Blog by Jason Ganz, DBT Labs, April 2025 <https://docs.getdbt.com/blog/introducing-dbt-mcp-server?version=1.12>
5. “[Just Use Postgres for Everything](#)”, Stephan Schmidt, updated, December 13, 2025
6. [Beginner's Guide to contributing to Airflow - Kalluri](#)
7. “Query Tools Criteria”, LinkedIn, Pradeep Kalluri, Jan 2026
8. Minio AIStor Documentation, <https://docs.min.io/aistor/>
9. Kafka documentation: <https://kafka.apache.org/43/getting-started/introduction/>
10. Why is Kafka replacing Zookeeper with KRaft: <https://www.confluent.io/learn/zookeeper-kafka/#apache-kafka-without-zookeeper-introduction-to-kraft>

References & Related Readings – page #2

1. KRaft vs Zookeeper: <https://kafka.apache.org/42/getting-started/zk2kraft/>
2. “Modern Data Engineering Meets MLOps: Building the Backbone of AI-Driven Products”, Anchit Gupta, Learn Data Now, via Medium, April 2026, [Link](#)
3. “The evolution of cloud-native data platforms for insurance analytics: A Technical Review”, A.R. Malipeddi, WJAETS, May 2025, [Link](#)
4. “Dashboards Are Dead. The Future Is Decision Execution”, Cloud with Azeem, via Medium, July 2026, [Link](#)
5. “Data Engineering System Design: Building a CDC-Based Pipeline from OLTP to a Lakehouse”, Sarath Sagi, Medium, Dec 2025, [Link](#).
6. “The next evolution of Data Engineering: From Data Pipelines to Context Pipelines”, Think Data, Medium, June 2026, [Link](#).
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8. “True enterprise sovereignty is more approachable than ever, thanks to K8s-powered cloud-neutral PostgreSQL”, EnterpriseDB, TheNewStack, April 2026. [Link](#).
9. “The Data Engineering: How Modern Data Infrastructure is Powering the AI Revolution”, Nithin Narla, Towards AI, Medium, April 2026, [Link](#).
10. “The Real Reason Data Teams Are Quietly Replacing Their Data Warehouses”, Saurav Singh, Medium, May 2026, [Link](#).